



UNIVERSITY OF MORATUWA
Faculty of Information Technology
B.Sc. (Hons) in Information Technology
B.Sc. (Hons) in Information Technology and Management
B.Sc. (Hons) in Artificial Intelligence
Level 2 – Semester 1 Examination

IN 2211 – OBJECT-ORIENTED ANALYSIS AND DESIGN

Time Allowed: 3 hours

November 2023

INSTRUCTIONS TO CANDIDATES

1. This paper contains 4 questions on 5 Pages, including this page.
2. The total marks obtainable for this examination is 100.
3. The marks assigned for each question are included in square brackets.
4. This examination accounts for 70% of the module assessment.
5. This is a closed-book examination.
6. Start answering each question using a new page.
7. Answer ALL questions.

ADDITIONAL MATERIAL

None.

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Question 01 [35 marks]

- a) Use case diagrams serve the purpose of analyzing and specifying functional requirements. They include actors, use cases, and depict relationships between actors and use cases.
- I. Briefly explain the difference between the *Include* and *Extend* relationships.
[4 marks]
 - II. Briefly explain the difference between *generalization* and *association* relationships, using an example to illustrate your explanation.
[4 marks]
- b) Consider the below scenario of an online food ordering system in the 'CravingFood' restaurant.

A customer can log into the system using existing account credentials or create a new account if it is the first time. Customers can use the filtering option to narrow down his/her selections based on food type, price range, and customer ratings. Customers can select and add food items to a virtual cart. Also, if required customers can customize each food item while adding it to the cart. A customer can customize his/her order by specifying preferences like spice level, additional toppings, and special instructions. Once the customer is satisfied with the food items added to the cart, he/she can place the order by making the relevant payment. If a customer has a discount coupon, then it can be entered at the time of placing the order to get a deduction in payment.

The order coordinator retrieves the order from the system and assigns it to a driver to make the delivery. The driver delivers the order to the customer.

- I. Draw the UML Use Case Diagram to visualize the above scenario.
[10 marks]
- II. Consider the below scenario of "Assign drivers" use case.

The order coordinator can request a list of drivers from the system, and he/she selects a driver from that list. Based on the selection, the system checks the driver's availability. This is repeated until a driver has been selected (until an available driver is found). Once the driver is found, the order coordinator assigns the driver using the system, and the system notifies the relevant driver about the delivery.

Draw the UML Sequence Diagram for the above "Assign drivers" use case.

[10 marks]

Continued...

- III. Write a use case description for the "Assign drivers" use case described in (b) II.

[7 marks]

Question 02 [25 marks]

- a) Briefly explain the following terms with suitable examples.
- I. Class
 - II. Object

[2 x 2 marks]

- b) Briefly explain *Composition* and *Aggregation* using suitable examples.

[2 x 2 marks]

- c) Consider the following description of a university.

A lecturer teaches subject modules in which students can enroll. A lecturer has a name, address, phone number, email address, and salary. A student also has a name, address, phone number, and email address, but no salary. A student, however, has a grade point average (GPA) (of the final grades of his/her subject modules). A subject module has a name and a number. When a student is enrolled in a subject, the marks for this enrollment are recorded and the current grades as well as the grade point average can be obtained from the enrollment. From a student, one can obtain a list of subject modules he or she is enrolled in. Each subject has at least one and at most two lecturers to teach the subject. There are two types of subjects: compulsory and elective. From a compulsory subject, students cannot withdraw. From an elective subject, they can.

Draw a UML Class Diagram for this university. Add attributes and methods when necessary. Visibility modifiers are not required.

[10 marks]

- d) Draw the Object Diagram for the relationship between the "Student", "Lecturer" and "Subject" classes.

[7 marks]

Question 3 [20 marks]

- a) Briefly elaborate on why software development is inherently seen as a complex task by providing two (2) reasons.

[2 x 2 marks]

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- b) Consider the following description.

The online printing service allows customers to print posters or flyers. The customer should be able to select a type of product (poster or flyer), a desired quantity, and a paper type. Finally, the customer must provide a PDF file containing the desired content.

For the customer to be able to place an order, he or she must have an account. Furthermore, his or her address and credit card number can be linked to the account, which is required information when placing an order. Once the online order is placed, the order processing starts.

Once an order has been finalized, four parties are involved in processing it: Online Sales, Accounting, Shipping, and Printing. Online Sales sends the order to Print, where the associated PDF file is inspected. If the file is not suitable, a new file is requested from the customer. Once the file is suitable, Accounting is informed to charge the credit card. While this is done, Printing carries out the actual printing and sends the result to Shipping. When Shipping has both received the printed products and confirmation from accounting that payment was successful, the products are shipped to the customer.

- I. Briefly explain the difference between *fork* and *join* in a UML Activity Diagram using a suitable example(s).

[6 marks]

- II. Draw the UML Activity Diagram for processing an order.

[10 marks]

Question 04 [20 marks]

- a) Explain the difference between the UML *Package Diagram* and the UML *Component Diagram*.

[6 marks]

- b) Explain the difference between the provided interface and the required interface in the UML *Component Diagram*.

[6 marks]

- c) Consider the below description regarding an online food ordering system.

The food ordering system includes three components: product management, customer management, and order processing. The product management component manages details of the food items (e.g., name, quantity, unit price, ingredients). The customer management component handles customer details (e.g., name, delivery address, credit card details) and order history.

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Order processing components manage the ordering and checkout process. The order processing component connects with the customer management component to receive customer details like the customer's delivery address. Similarly, the order processing system connects with the product management component to get product details. The customer, product, and order details are shown through a user interface to the user.

Draw the UML Component Diagram for the scenario given above.

[8 marks]

End of the Paper.